

5.2.1 Definite Integral Worksheet

1. Let A be the area of the region that lies under the graph and above the x -axis of $f(x) = \ln(x)$ between $x=1$ & $x=e$. Using right endpoints, find an expression for A as a limit.

2. Use the definition of the definite integral to express $\int_{-2}^2 \frac{\sin^2(x)}{x+3} dx$ as a limit.

3. The area between the x -axis and the graph of $f(x) = \frac{(x+1)^2}{\ln(x)}$ on the interval $[3,15]$ can be written as a limit. Fill in the missing information in this limit so that the only variables appearing are n and i . You do not need to evaluate this limit.

$$\int_3^{15} \frac{(x+1)^2}{\ln(x)} dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\right]$$

4. Determine a region whose area is equal to the limit $\lim_{n \rightarrow \infty} \sum_{i=1}^n \left[\sqrt{5 + \frac{2i}{n}} \cdot \frac{2}{n} \right]$. You do not need to evaluate the limit.
5. At time t seconds, the velocity of an object is given by the equation $v(t) = t^3 \text{ ft/sec}$. The distance in feet traveled by this object from $t = 12$ to $t = 20$ can be written as a limit of Riemann sums in some different ways. Use your knowledge of Riemann sums to set up the following limits but do not evaluate them.
- a) Using a limit of right Riemann sums,
- b) Using a limit of left Riemann sums,
- c) Using a limit of midpoint Riemann sums,
6. Evaluate the limit $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{(3k + 2n)^2}{n^3}$. Use proper notation throughout.

7. Use the definition of the definite integral to write the integral $\int_2^4 (x - x^2) dx$ as a limit. Then evaluate this limit.

8. Use the definition of the integral to evaluate $\int_1^5 (x^2 + 3x + 2) dx$.